Reading Group Comparisons: Categorical Outcomes and Mean Outcomes

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Chapter box

Central question: How can a reader understand and compare group results when the outcome is categorical or quantitative?
Focal concepts: Denominators, conditional proportions, grouped distributions, estimands, omnibus questions, single and multiple statistical strands, simultaneous uncertainty, and design-bounded claims.
Main evidence: One compact observational comparison from the UCI student record and one transparently hypothetical teaching extension based on the documented Baumann reading-study record. The first pair has one estimand and one statistical strand. The second has one integrated question and two subordinate contrast estimands.
Scope boundary: The main route covers ordinary two-way tables and one-way group-mean comparisons, including the basic ANOVA model, sums of squares, degrees of freedom, and F calculation. Sparse tables, extended effect scales, full ANOVA derivations, robust alternatives, blocking, and software implementation are optional depth.
Use hierarchy: Reading published comparisons is the primary use. Quality evaluation is the distinct secondary use after reading. The six-stage construction workflow is the auxiliary use for conducting research.

Chapter 1 supplies the shared argument-based vocabulary; Chapter 1a is not required. The calculations build on the means, variances, standard errors, intervals, and tests developed in Chapters 2–3. Brief reminders are included, but introductory statistics and basic algebra are useful preparation. The public route requires no programming or statistical software.

1 Begin with the Comparison Question

Consider two statements that could appear in a report:

> Students intending higher education were more likely to pass.
>
> The instructional methods differed significantly.

Both may sound informative, yet neither is a complete answer. The first omits the denominator, numerical difference, source population, and observational boundary. The second omits the outcome, the groups, the comparison that matters, and the size and uncertainty of the difference.

A reader can first interpret either statement by building a **comparison record**; subsequent quality evaluation may show that its wording needs repair:

1. **Unit and analytic sample:** What does one observation represent, and which observations entered the result?

2. **Group or assigned condition:** Which groups are being compared, and were they observed or assigned?
3. **Outcome:** Is it a named event or a numerical scale with units?
4. **Estimand:** Is the quantity sought an event-probability difference, an equality pattern, a mean difference, or another declared contrast?
5. **Design-supported scope:** Which people, settings, processes, or assignments can the answer describe?

Table 1
A reading-first orientation to group comparisons.

Outcome form	First evidence to inspect	Typical estimand	First effect estimate
Categorical	Counts and conditional proportions	Difference in event probabilities or a whole-table association	Proportion difference for a named event and direction
Quantitative	Raw grouped observations and group summaries	Difference or contrast among group means	Mean difference or prespecified linear contrast

This table is an orientation, not a mechanical procedure selector. The design and the intended claim still determine whether a repeated-sample or causal interpretation is justified.

An **estimand** is the precisely specified quantity the study seeks to learn; an **estimate** is the numerical result obtained from the observed evidence for that estimand. Several tables, intervals, tests, or sensitivity checks may describe one estimand without creating several arguments. Several estimands become **statistical strands** only when all are subordinate to one integrated answer and none is owed a separate question-facing conclusion. If a report interprets a contrast separately, that conclusion belongs to its own question–answer pair and argument.

The remainder of the main route follows Figure 1 twice. The UCI record supplies a categorical observational comparison. The Baumann-inspired teaching contract supplies a quantitative randomized comparison. They share a reading discipline, but they do not share evidence or one combined argument.

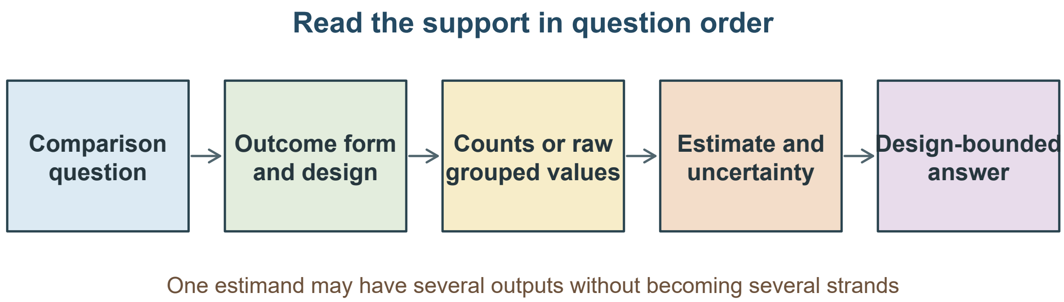

Figure 1: A reading route from the comparison question to a design-bounded answer.

2 Read a Categorical Group Comparison

2.1 Start with the denominator

The first question is:

> Among students recorded in two Portuguese secondary schools, how did the observed passing proportion differ between students who did and did not report an intention to pursue higher education?

The UCI Student Performance record contains 395 students from two schools (Cortez, 2008), distributed under the Creative Commons Attribution 4.0 licence. One row represents one recorded student. Higher-education intention was observed, not assigned. The values embedded here are unchanged; the adaptation selects and renames the analysed columns and derives the pass indicator from the documented cut point of 10 on the 0–20 scale.

The source documentation defines final grade on a 0–20 scale and marks the file as having no missing values. It does not, however, explain an unusual joint pattern: 38 records have final grade 0 and absences 0, although 25 of those records have a positive second-period grade. A zero may be a genuine lowest grade, but the accessible record does not establish whether every zero has the same meaning. The primary analysis therefore keeps the published numeric coding exactly as recorded and labels the unresolved pattern; a later deletion analysis is only a diagnostic sensitivity check.

The question asks for the proportion passing **within each intention group**. The group totals are therefore the denominators:

$$\hat{p}_g = \frac{\text{number passing in group } g}{\text{number recorded in group } g}.$$

Table 2

Pass status and within-group passing proportions from grades as recorded in the UCI record

Higher-education intention	Did not pass	Passed	Group total	Passing proportion
No	13	7	20	0.350
Yes	117	258	375	0.688

The group that reported an intention to pursue higher education was much larger: 375 students, compared with 20 who did not. Counts alone therefore mislead. Within the two groups, the passing proportions were .688 and .350.

Column percentages would answer a different question: among students who passed or did not pass, what proportion reported higher-education intention? Neither row nor column percentages are universally correct. The research question chooses the denominator.

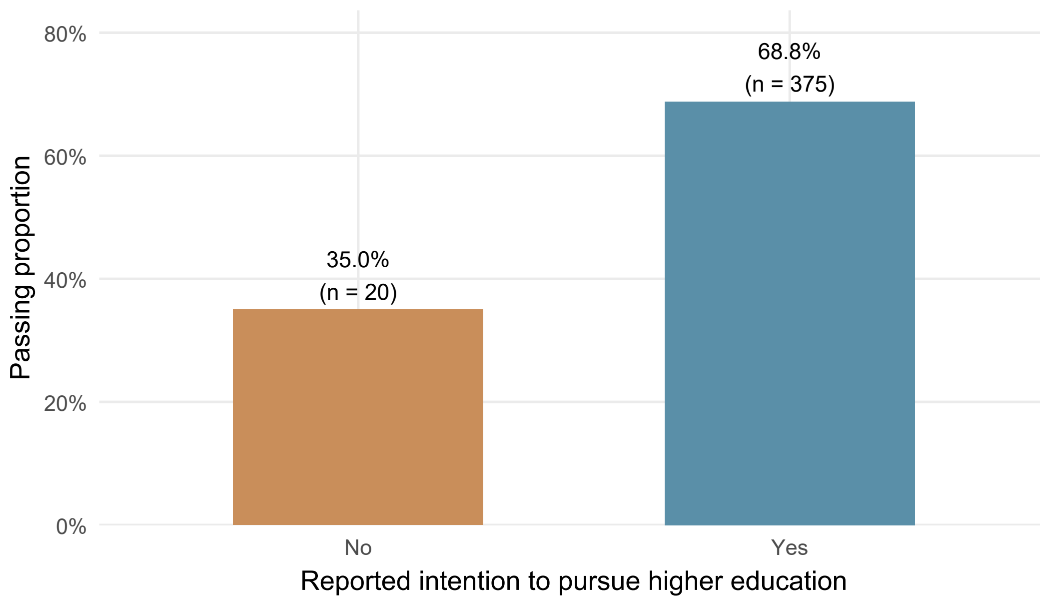

Figure 2: Within-group passing proportions derived from grades as recorded in the UCI record.

For the declared event and direction, the primary descriptive effect is

$$\widehat{\Delta} = \hat{p}_{\text{yes}} - \hat{p}_{\text{no}} = .688 - .350 = .338.$$

Thus, among these records, the passing proportion was 33.8 percentage points higher in the intention-yes group. This is already a complete **descriptive** answer for the observed table. No p-value is needed to know that realized difference.

Other scales answer other questions. **Odds** are $p/(1-p)$: passing probability .75 gives odds $.75/.25 = 3$, or three passes per nonpass. A probability ratio compares probabilities; an **odds ratio** compares odds, not probabilities. Appendix A develops these scales. The proportion difference remains primary because percentage points directly answer our question.

2.2 Read evidence against an independence pattern

Inference adds a question that description alone does not answer:

> Would a discrepancy of this size be unusual under a declared no-association model or assignment process?

For a two-way table, **independence** means that the modeled distribution of pass status is the same across intention groups. Under that restriction, the expected count in row r and column c is

$$E_{rc} = \frac{(\text{row total})(\text{column total})}{n}.$$

For example, the no-intention group contains 20 students and the table contains 265 passes. The expected no-intention/pass count under independence is $(20)(265)/395 = 13.42$, compared with 7 observed passes. The Pearson statistic collects the cell discrepancies:

$$X^2 = \sum \frac{(O-E)^2}{E}.$$

Here X^2 names the calculated Pearson statistic; χ^2 names its chi-square reference distribution. Reports often use χ^2 for the reported statistic too; these are notation conventions, not different tests.

Here O and E are an observed and expected cell count, and $\sum$ means add the contributions across all cells. For the no-intention/pass cell, the contribution is $(7-13.4177)^2/13.4177 = 3.0696$. The other three contributions are 6.2573, 0.1637, and 0.3337, giving $X^2 = 9.8243$ after rounding. The reference degrees of freedom are $(R-1)(C-1)$ for R rows and C columns: here $(2-1)(2-1) = 1$. Once both margins are fixed, specifying one cell fixes the remaining cells; they cannot vary independently.

It is an omnibus statistic for the whole independence restriction. It is not an effect size, a causal estimate, or a locator that by itself explains which cells produced the pattern.

2.3 Make the model-based uncertainty checkable

The recorded-table question has already been answered descriptively. For an explicitly conditional illustration of inference, consider comparable records from the same stipulated process. Write $\pi_g = P(Y = 1 \mid G = g)$ for the passing probability within intention group g, where $Y = 1$ means pass. The model estimand is $\Delta = \pi_{\text{yes}} - \pi_{\text{no}}$; its estimate is the observed difference $\widehat{\Delta} = .338$. That model quantity differs from the as-recorded 395-record description. The model calculation is contextual output here, not uncertainty about the already known table or an independently established conclusion about other schools.

Under independent binomial sampling within the two groups, estimate the standard error by adding the variances of their estimated proportions:

$$\widehat{SE}(\widehat{\Delta}) = \sqrt{\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \frac{\widehat{p}_0(1-\widehat{p}_0)}{n_0}} = \sqrt{\frac{.688(.312)}{375} + \frac{.350(.650)}{20}} = .1093.$$

Subscripts 1 and 0 denote the yes and no groups. An illustrative Wald interval is $.338 \pm 1.96(.1093) = [.124, .552]$. The small 20-student group makes this normal approximation fragile; a score-based interval is generally preferable for a substantive analysis. The calculation is retained here to make the standard-error mechanism visible, not as a recommendation to use Wald intervals routinely.

The no-association test asks a different numerical question: if both groups shared one passing probability, how surprising would the discrepancy be? Its pooled estimate is $\widehat{p} = 265/395 = .6709$, so its null standard error is $\sqrt{\widehat{p}(1-\widehat{p})(1/375+1/20)} = .1078$. The resulting $z = .338/.1078 = 3.1344$ has $z^2 = 9.8243$, the uncorrected Pearson statistic for this 2×2 table. The pooled null SE is not the unpooled estimation SE used for the displayed interval. These are outputs about the same directed model difference under different restrictions, not new statistical strands.

Table 3

Categorical estimate, interval, and omnibus evidence derived from grades as recorded

Result	Value
Passing proportion, intention yes	0.688
Passing proportion, intention no	0.350
Difference, yes minus no	0.338
Large-sample 95% CI for difference	[0.124, 0.552]
Pearson chi-square	9.82
Degrees of freedom	1
p value	.002

Table 3 keeps four tasks separate:

- The **proportion difference** gives direction and magnitude on the selected event scale.
- The **interval** displays values compatible with a hypothetical record-generating model. The displayed Wald interval is illustrative and should be treated cautiously because the no-intention group has only 20 students.
- The **chi-square result**, chi-square with 1 degree of freedom = 9.82, $p = .002$, assesses the whole independence restriction under its reference conditions.
- The **question wording** determines whether the mode is descriptive, associational, predictive, or causal. The **design and justification** determine which claim is warranted.

All four expected counts exceed 5 in this table, but that fact is not a universal license to trust the reference distribution. The analysis also requires a defensible account of dependence, selection, clustering, measurement, and the stochastic process to which the p-value refers. Students share schools and perhaps classrooms, and the source is not a probability sample of a broad population.

2.3.1 Check the unexplained zero-grade cluster

If the 38 final-grade-zero records are omitted solely as a diagnostic, 357 records remain. The passing proportions become .752 in the intention-yes group (258/343) and .500 in the intention-no group (7/14), a difference of .252. The illustrative Wald interval becomes [-.014, .518]. Pearson's $X^2(1) = 4.47$, $p = .034$, but its smallest expected count is only 3.61; Fisher's two-sided exact test gives $p = .055$. Thus, even this simple coding decision materially changes the numerical evidence.

This is not permission to delete the zeros. Doing so changes the analytic set and the finite-record quantity being described, and it requires a defensible record-definition or missingness rule that is not located in the source documentation. The comparison based on all 395 rows remains the primary source-coded result; the diagnostic shows why its meaning and process-level extension require caution.

! Bounded categorical answer

Among the 395 recorded students, the observed passing proportions were 68.8% for those reporting an intention to pursue higher education and 35.0% for those not reporting it, a difference of 33.8 percentage points. Under an illustrative independent-record model for comparable records from the same process, a large-sample 95% interval was 12.4 to 55.2 percentage points and the independence statistic was chi-square with 1 degree of freedom $= 9.82$, $p = .002$. A diagnostic omission of the 38 unexplained final-grade-zero records reduced the difference to 25.2 percentage points and made the inferential conclusion sensitive to the procedure (Pearson $p = .034$; Fisher $p = .055$). The all-row comparison is exploratory, local, and observational. It does not show that intending higher education caused passing or support transport to students outside the documented setting and selection process, and the accessible materials do not settle the meaning of the zero-grade cluster.

3 Read a Mean Comparison with One Integrated Answer

3.1 Define the integrated comparison question

The second example deliberately separates a **source record** from a **chapter-created teaching contract**. Baumann and colleagues assigned 68 fourth-grade students in one rural Midwestern school: 22 to directed reading activity (DRA), 23 to directed reading-thinking activity (DRTA), and 23 to think-aloud instruction (Baumann et al., 1992, p. 147). One student in each innovative condition had an incomplete outcome because of illness or transfer. The public *Baumann* record in the *carData* package therefore contains 66 observed outcomes, 22 per group (Fox et al., 2022; Fox & Weisberg, 2019).

The retained *carData* record contains two pretest variables and three quantitative post-test variables. The source study also administered a further qualitative interview post-test to a subset of pupils; that material is not part of this teaching record. This chapter selects the 0–16 error-detection outcome and uses an **unadjusted one-way model**, rather than reproducing the source's pretest-adjusted analysis. The uncertainty calculations are model-based, not finite-sample randomization intervals. The package cannot recover the two unobserved scores. These are limits of this teaching reanalysis, not details supplied by an ANOVA result.

The chapter does **not** claim that the source investigators prespecified the question, contrasts, or decision rule below. Instead, imagine a future replication in this setting whose teaching contract declares, before its outcomes are inspected:

> Does a two-format instructional package meet its integrated success criterion: both DRTA assignment and think-aloud assignment improve mean error-detection score relative to DRA assignment?

For this hypothetical contract, *improve* means a positive mean difference. There are two subordinate estimands:

1. mean error-detection score under DRTA assignment minus mean score under DRA assignment; and
2. mean score under think-aloud assignment minus mean score under DRA assignment.

For this hypothetical replication, the intended means refer to outcomes under each assignment for comparable eligible pupils in the stipulated setting. The inferential calculations use an independent-pupil outcome model, not a claim that this school represents all fourth-graders. All assignees belong in that intended comparison; if only complete outcomes are analysed, linking observed means to those intended means requires an explicit missingness justification. The chapter reports that limitation rather than quietly redefining the estimands as effects among only the observed pupils.

The declared rule requires the lower endpoint of **each** two-sided 95% Bonferroni simultaneous interval for those two comparisons to exceed zero. Zero is a directional benchmark, not a threshold for educational importance. The criterion is prospective only inside this explicitly hypothetical replication. Applying it to an already available source record is a post hoc, secondary teaching reanalysis.

The two estimands count as statistical strands because both are needed for the one package-level answer and neither receives its own recommendation. If a report instead concludes separately about DRTA or think-aloud, it owes a separate question–answer pair and argument for that conclusion.

3.2 See grouped evidence and read ANOVA

The assignment record and observed outcomes come before a test.

Table 4
Source assignment record and observed error-detection summaries

Group	Assigned	Observed	Mean	SD	Range
DRA	22	22	6.68	2.77	2–12
DRTA	23	22	9.77	2.72	5–14
Think-aloud	23	22	7.77	3.93	1–15

The assigned counts and observed counts are not interchangeable. The 66 complete records support direct description of observed outcomes. Extending that analysis to an all-assigned estimand requires a missingness justification; ANOVA cannot supply one.

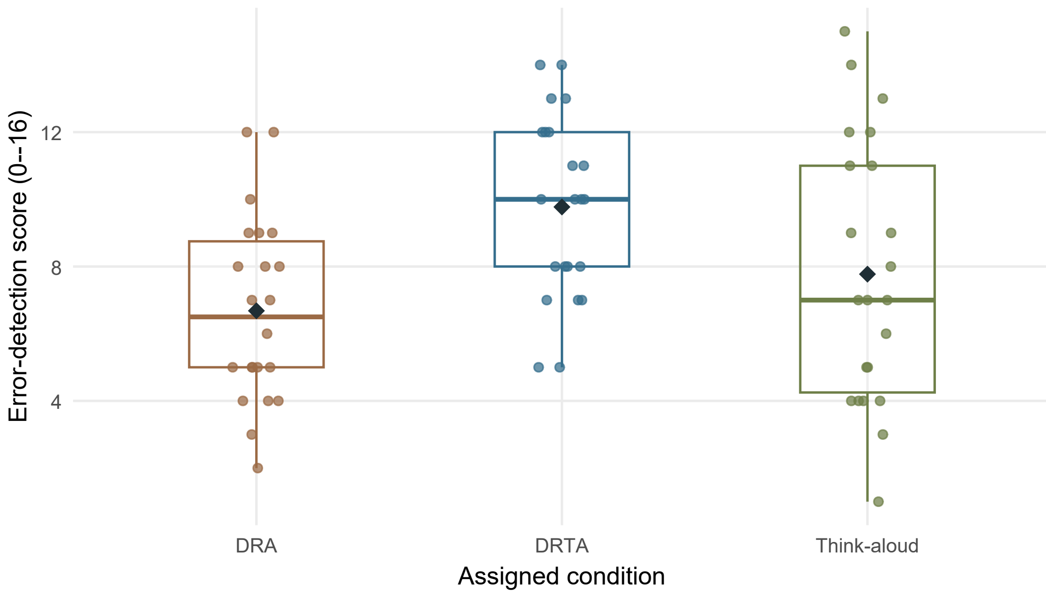

Figure 3: Observed error-detection scores by assigned instructional condition.

Before seeing an ANOVA table, Figure 3 and Table 4 establish four facts:

- the group centers differ;
- the observations overlap;
- within-group variation is substantial; and
- the think-aloud group is more dispersed than the other two groups.

The graph is evidence about the observed distributions. It does not prove a model assumption or a causal claim.

3.2.1 Read the omnibus ANOVA question

One-way ANOVA represents each observed score as

$$\text{observed score} = \text{fitted group mean} + \text{within-group residual}.$$

The corresponding **group-mean model** is $Y_{ij} = \mu_j + \varepsilon_{ij}$: Y_{ij} is pupil i's score in group j, μ_j is that condition's model mean, and ε_{ij} is the pupil's deviation from it. Assume mean-zero errors, a common variance σ^2, and independent pupils for this working model. The fitted mean is the observed group mean $\bar{Y}_j$; a residual is $Y_{ij} - \bar{Y}_j$. A fitted residual is not the same as the unknown model error ε_{ij}.

3.2.2 From deviations to sums of squares

The **grand mean** $\bar{Y}$ is the mean of all $n = 66$ observed scores; it is 8.0758. Each deviation from it splits into two pieces:

$$Y_{ij} - \bar{Y} = (\bar{Y}_j - \bar{Y}) + (Y_{ij} - \bar{Y}_j).$$

For a DRA pupil scoring 5, this is $5-8.0758 = (6.6818-8.0758)+(5-6.6818) = -1.3940-1.6818$. The first piece locates the group mean; the second locates the pupil within that group. Square and add across pupils and groups to obtain

$$\underbrace{\sum_j \sum_i (Y_{ij} - \bar{Y})^2}_{SS_T} = \underbrace{\sum_j n_j(\bar{Y}_j - \bar{Y})^2}_{SS_B} + \underbrace{\sum_j \sum_i (Y_{ij} - \bar{Y}_j)^2}_{SS_W}.$$

The double summation means sum over pupils within each group and then over groups. SS_T, SS_B, and SS_W are total, between-group, and within-group sums of squares. The extra cross-products vanish because residuals within each group sum to zero. For example, DRA contributes $22(6.681818 - 8.075758)^2 = 42.7475$ to SS_B. Across all three groups, $SS_B = 108.1212$ and $SS_W = 640.5000$, so $SS_T = 748.6212$. These numbers measure variation in squared score units, not mean differences.

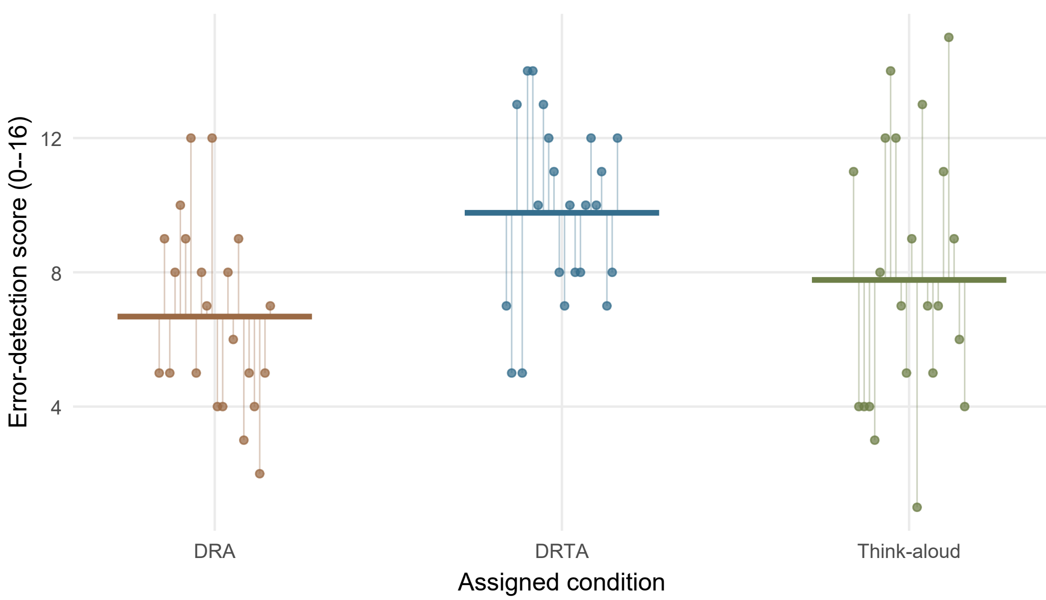

Figure 4: Fitted group means and within-group residuals in the Baumann data.

3.2.3 From mean squares to the omnibus test

Divide each sum of squares by its degrees of freedom to obtain a **mean square**. There are $J = 3$ fitted group means, so between-group degrees of freedom are $J - 1 = 2$ and residual degrees of freedom are $n - J = 63$. The three within-group zero-sum constraints leave $66 - 3 = 63$ free residual pieces.

$$MS_B = \frac{108.1212}{2} = 54.0606, \qquad MS_W = \frac{640.5000}{63} = 10.1667,$$
$$F = \frac{MS_B}{MS_W} = \frac{54.0606}{10.1667} = 5.3174.$$

MS_W, also called the mean square error or MSE, estimates the common error variance. Under equal model means, both mean squares estimate that variance, so an F ratio much larger than 1 indicates an unusually large between-group pattern relative to within-group noise. The omnibus null is

$$H_0 : \mu_{\text{DRA}} = \mu_{\text{DRTA}} = \mu_{\text{Think-aloud}}.$$

It is one joint question. The numerator degrees of freedom describe the number of independent group restrictions; the denominator degrees of freedom describe the residual information used to estimate within-group variation.

Under independent Normal errors with common variance, the null reference is exactly an F distribution with $J - 1$ and $n - J$ degrees of freedom, conditional on the group assignments. Without those error conditions it is an approximation whose suitability needs justification. Random assignment supports the assignment comparison; it does not prove Normal errors, equal variances, or the exact finite-sample F reference.

Table 5

One-way ANOVA for observed error-detection outcomes

Source	df	Sum of squares	Mean square	F	p
Between groups	2	108.121	54.061	5.317	0.007
Within groups	63	640.500	10.167		
Total	65	748.621			

The omnibus result, $F(2, 63) = 5.32$, $p = .007$, indicates that the equal-means restriction is difficult to reconcile with the declared pooled model. It does **not** identify which comparison matters, show that every pair differs, or give the size of a meaningful comparison.

Model conditions are questions about how the evidence was produced:

- Were units independent under the assignment and classroom setting?
- Did the score have the same meaning in every condition?
- Is a common within-group variance a useful working model?
- Are unusual observations or distributional shapes influential?
- Does the treatment of missing outcomes align with the all-assigned estimand?

A plot can reveal a possible problem. It cannot certify an assumption. For these observed outcomes, Welch's unequal-variance sensitivity result is $F(2, 41.13) = 6.99$, $p = .002$. It relaxes the common-variance condition; it does not repair missingness, dependence, outcome selection, or transport. Appendix B gives a small-data reconstruction and comparison-family detail.

3.3 Read two statistical strands and one integrated answer

The omnibus result is useful context, but it is not a third strand and cannot answer the package question. It can reject equal means when only one innovative condition differs from DRA. The integrated rule instead requires adequate evidence for **both** declared contrasts.

Table 6
Omnibus, statistical-strand, and separate-pair roles.

Result family	Role in this teaching case	Argument status
Omnibus ANOVA	Assesses the broader equal-means restriction	Contextual output; neither a strand nor the package answer
DRTA minus DRA	Supplies a complete-record estimate and simultaneous model interval relevant to estimand 1	Statistical strand 1
Think-aloud minus DRA	Supplies a complete-record estimate and simultaneous model interval relevant to estimand 2	Statistical strand 2
Other pairwise comparisons	Answer separately declared exploratory questions	Not strands of this integrated pair

3.3.1 Start with differences, then generalize to a contrast

The next calculations use the 66 complete records. They estimate differences among the observed group means. Connecting those values to the two all-assigned estimands declared above requires the missingness bridge that the source record cannot provide by itself.

The group means are 6.681818 (DRA), 9.772727 (DRTA), and 7.772727 (think-aloud). The two required estimates are therefore

$$\widehat{L}_1 = 9.772727 - 6.681818 = 3.090909, \qquad \widehat{L}_2 = 7.772727 - 6.681818 = 1.090909.$$

In the fixed order DRA, DRTA, think-aloud, these differences use weights $(-1, 1, 0)$ and $(-1, 0, 1)$. For the first, multiply the DRA mean by -1, the DRTA mean by 1, and the think-aloud mean by 0, then add. Each weight set sums to zero, so it compares means rather than their overall level.

A **linear contrast** generalizes this operation. With c_j denoting the chosen weight on group j,

$$L = \sum_{j=1}^{J} c_j \mu_j, \qquad \sum_{j=1}^{J} c_j = 0, \qquad \widehat{L} = \sum_{j=1}^{J} c_j \bar{Y}_j.$$

L is the estimand; $\widehat{L}$ is its estimate. Adding the same constant to every group mean leaves the contrast unchanged because the weights sum to zero. Changing the weights changes the comparison and may change the question–answer pair. In particular, averaging the two innovative conditions against DRA uses $(-1, 1/2, 1/2)$; it cannot answer whether **both** comparisons meet the declared rule.

3.3.2 Calculate uncertainty for the declared two-comparison family

Under the independent common-variance model, the estimated contrast SE is

$$\widehat{SE}(\widehat{L}) = \sqrt{MS_W \sum_{j=1}^{J} \frac{c_j^2}{n_j}}.$$

The squared weights appear because the variance of a weighted mean is multiplied by the square of its weight. For either declared contrast, $n_j = 22$ and only the two nonzero weights contribute: $\widehat{SE} = \sqrt{10.166667(1/22 + 1/22)} = 0.961375$ points. This is the uncertainty of a mean difference, not the SD of individual scores.

Bonferroni calibration protects a declared finite family of comparisons. For $m = 2$ two-sided intervals with family confidence at least 95%, allocate $.05/2 = .025$ noncoverage to each interval, split equally as $.025/2 = .0125$ in each tail. Thus $1 - .0125 = .9875$ lies to the left of the upper critical point. The resulting critical value is $t^* = t_{1-.05/(2m),63} = t_{.9875,63} = 2.296237$ (use **2.296** on a calculator). The subscript .9875 identifies the proportion to the left of that point in the t distribution; 63 is its degrees of freedom. Thus

$$\widehat{L}_k \pm t^* \widehat{SE}(\widehat{L}_k), \qquad k = 1, 2.$$

The margin is $2.296237(.961375) = 2.207545$ points. Strand 1 gives $3.090909 \pm 2.207545 = [0.883364, 5.298454]$; strand 2 gives $1.090909 \pm 2.207545 = [-1.116636, 3.298454]$. Round only at the end. The family statement concerns repeated-use coverage of **both** estimands, not a 95% probability for either fixed estimand after observing this pair of intervals. It also does not repair missingness or justify the outcome model.

Ordinary pointwise 95% intervals use a smaller critical value and do not provide this same simultaneous guarantee. Bonferroni does not require the two contrasts to be independent; their common DRA mean in fact makes them dependent. Other family-calibration procedures are possible. This teaching contract deliberately specifies Bonferroni; it is not a universal requirement for every claim requiring two positive differences. This family covers only the **two declared contrasts on the selected error-detection outcome**; it does not adjust across every outcome or analysis in the source study. Appendix B compares alternative families.

Table 7

Two complete-record DRA-referenced statistical strands under the hypothetical teaching contract

Strand	Estimate (complete records)	95% Bonferroni interval	Relation to zero	Conclusion
1: DRTA minus DRA	3.09	[0.88, 5.30]	Lower bound is above zero	Positive
2: Think-aloud minus DRA	1.09	[-1.12, 3.30]	Interval includes zero	Inconclusive

The DRTA-minus-DRA estimate is 3.09 points, with simultaneous interval [0.88, 5.30]. Its lower bound exceeds zero. The think-aloud-minus-DRA estimate is 1.09 points, with interval [-1.12, 3.30]. That interval remains compatible with both no improvement and a positive improvement. It is therefore **inconclusive** for the declared directional criterion. An interval including zero is not proof of equality.

3.3.3 Stress-test the two unavailable outcomes

The outcome scale runs from 0 to 16. Without pretending to know the two unavailable scores, let the missing DRTA score and the missing think-aloud score each take every integer value on that scale. The DRA group has all 22 scores, with sum 147. The observed DRTA and think-aloud sums are 215 and 171. The resulting **completed-data difference-in-means estimates** must therefore fall within

$$\frac{215 + x_D}{23} - \frac{147}{22} \in [2.666, 3.362], \qquad \frac{171 + x_T}{23} - \frac{147}{22} \in [.753, 1.449],$$

where $x_D, x_T \in \{0, 1, \dots, 16\}$. These are bounds on what the arithmetic difference in completed group means could be, not identification bounds for the causal estimands.

As a second, explicitly model-dependent check, refit the same pooled one-way model and both Bonferroni intervals for all $17^2 = 289$ possible completions. The DRTA-minus-DRA lower endpoint remains above zero; its smallest value is .271. The think-aloud-minus-DRA interval includes zero in every completion: its largest lower endpoint is -.805 and its smallest upper endpoint is 2.996. Consequently, the chapter's conjunctive statistical criterion remains unmet for every bounded completion under this particular analysis rule.

This is a **bounded completed-data decision stress test**, not a formal missing-data analysis. It demonstrates that the teaching decision is stable across these scale-bounded completions, but it does not establish why outcomes were unavailable, identify all-assigned causal effects, repair outcome selection, or justify transport beyond the setting.

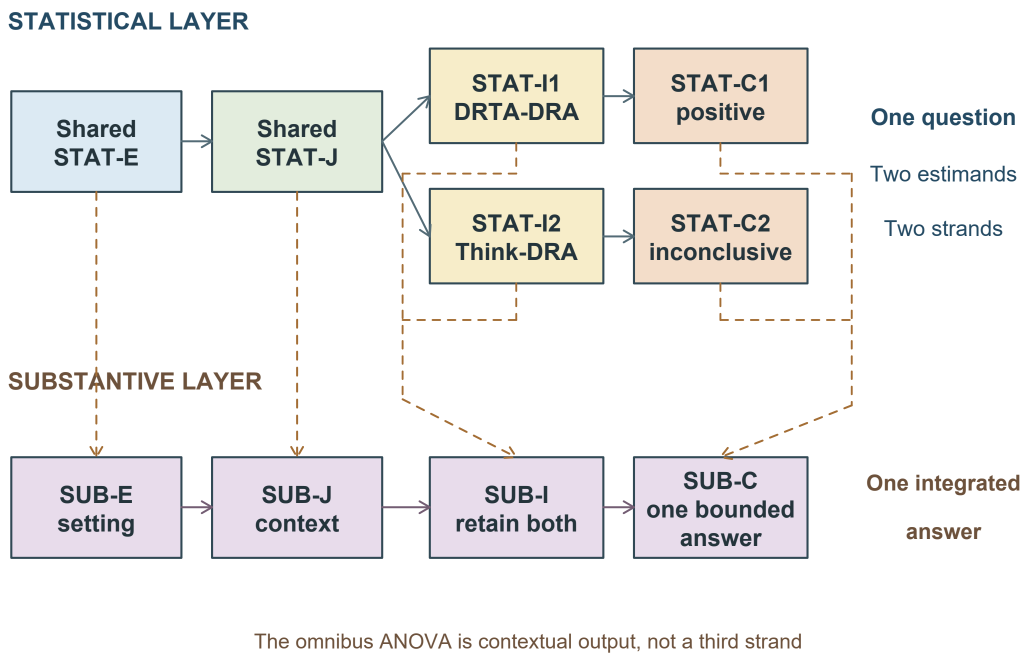

Figure 5: Two statistical strands supply one integrated teaching answer.

Figure 5 keeps three counting rules visible:

- several outputs for one estimand remain in one statistical strand;

- two estimands form strands here only because both are jointly required for one integrated answer and neither is owed a separate conclusion; and
- if either contrast is interpreted or recommended separately, it becomes a separate question–answer pair and argument.

The two-format statistical success criterion is **not met** in the complete-record analysis and remains unmet in all 289 scale-bounded completions under the same model and decision rule, so the integrated answer is **inconclusive**. This does not show that the package failed or that think-aloud has no effect. It says that the observed evidence does not meet the conjunctive rule. The statistically detectable omnibus result and a positive average of the two innovative conditions cannot replace the missing strand-level support.

> **!** Bounded integrated answer
>
> For the 66 complete records in this secondary teaching reanalysis, DRTA minus DRA was 3.09 error-detection points (two-sided 95% Bonferroni simultaneous interval [0.88, 5.30]), whereas think-aloud minus DRA was 1.09 points ([-1.12, 3.30]). The first comparison met the hypothetical positive-difference rule and the second was inconclusive. Consequently, the chapter-created two-format statistical success criterion was not met; the integrated substantive answer remains inconclusive. The same statistical criterion stayed unmet when each unavailable outcome was allowed to take every integer score from 0 through 16 under the same pooled model. That bounded completed-data stress test does not identify the all-assigned effects or replace a missingness analysis. This answer is post hoc relative to the source study even though the criterion is prospective inside the imagined replication. It is conditional on the pooled model and bounded completion rule, with missingness unresolved; it is limited to the selected outcome and documented setting, and is not a claim that either method failed or that effects transport broadly.

4 Read, Evaluate, and Construct the Arguments

4.1 Let the design bound the substantive answer

The two examples used different procedures, but the larger difference lies in their designs.

Table 8
How design bounds the two group-comparison claims.

Feature	UCI categorical pair	Baumann quantitative pair
Group variable	Observed intention	Randomly assigned instructional condition
Outcome	Recorded pass status	Error-detection score

Feature	UCI categorical pair	Baumann quantitative pair
Argument granularity	One estimand and one statistical strand	One hypothetical integrated question with two subordinate contrast estimands
Immediate estimate	One proportion difference	Two retained DRA-referenced mean differences
Strongest design-based language	Local association	Assignment comparison under stated conditions
Main reservations	Selection, confounding, shared setting, transport	Post hoc application of the teaching criterion, missing outcomes, selected outcome, model-based uncertainty, transport

A small p-value would not turn the UCI comparison into an experiment. A more complex model would not create random sampling. Likewise, random assignment in the Baumann study supports a comparison of assignment policies in its setting, but it does not by itself solve missing outcomes, define educational importance, or justify transport.

4.2 A short regression bridge

The same group means, contrasts, and omnibus restriction can also be represented using indicator-coded regression. That representation is developed with multiple regression, after simple-regression coefficients have been introduced. Re-encoding the same comparison does not strengthen its design.

Adding another factor creates adjustment, weighting, and possible interaction questions developed in the later regression chapters. Appendix B's blocking discussion stays narrower: it shows how blocking can refine one declared group-comparison estimand without introducing coordinated questions.

Chapter 1 treats one question–answer pair as **one argument with linked statistical and substantive layers**. Within either layer,

$$E + J \longrightarrow I \longrightarrow C.$$

Reading is the primary use. It asks what content performs each component, where that content can be inspected, and what answer the authors or teaching contract actually report. It does not yet decide whether the support is adequate.

Four documentary statuses keep location separate from quality: **stated in the paper**, **located in related materials**, **implicit but traceable**, and **not located**. A contradiction across sources is a separate conflict flag. The chapter-created questions and analyses below are labeled as such rather than misdescribed as documentary traces of the source investigators' choices.

4.3 Read the UCI single-strand argument

Table 9
Reading record for the UCI single-strand argument.

Component	Content identified during reading	Trace and documentary status
STAT-E	395 records; intention and pass coding; group counts and denominators; unexplained cluster of 38 final-grade zeros	UCI record and data dictionary: **located in related materials**; a shared meaning for all zeros is **not located**
STAT-J	Within-group denominators; declared independent-record approximation for the interval and chi-square reference; all-row coding kept primary and zero-row omission labeled diagnostic	Formula and assumptions are **stated in this chapter**; source-population and zero-grade processes are **not located**
STAT-I	Recorded-table difference .338; separate model illustration gives 95% interval [.124, .552] and $X^2(1) = 9.82$, $p = .002$; deletion diagnostic gives difference .252 with method-sensitive p-values	Chapter-created analysis, reproducible from the related data
STAT-C	A known 33.8-percentage-point difference in the all-row table; model uncertainty is conditional context, and the diagnostic exposes sensitivity to record definition	**Stated in this chapter**
SUB-E	Students recorded in two Portuguese secondary schools; observed intention; passing from final grade	UCI documentation: **located in related materials**
SUB-J	Observational grouping cannot by itself exclude confounding or establish transport; documentation does not settle the meaning of the zero-grade cluster	Design relation is **implicit but traceable** from the record; broader sampling and zero-grade warrants are **not located**
SUB-I	Interpret .338 as a local association, not an effect of intending	**Stated in this chapter**
SUB-C	The bounded categorical answer in Section 2	**Stated in this chapter**

The primary recorded-table estimand is followed through one statistical strand. The count table and proportional plot document it; the model interval and chi-square output show what an explicitly conditional inferential extension would add. They do not silently turn a known finite-record description into a population estimate. A report making a separate process-level conclusion would owe a corresponding question–answer pair and its justification.

4.4 Read the hypothetical two-strand argument

The source record supplies study context and outcomes. The integrated success criterion is supplied only by the chapter's hypothetical teaching contract. No row below attributes that criterion, its contrasts, or its Bonferroni rule to Baumann and colleagues.

Table 10
Statistical-layer reading record for the two-strand teaching argument.

Statistical component	Shared core	Strand 1: DRTA minus DRA	Strand 2: Think-aloud minus DRA
STAT-E	Assignment record; 68 assigned; 66 observed scores; one incomplete outcome in each innovative arm	DRA and DRTA observations	DRA and think-aloud observations
STAT-J	Assignment rationale; common score; independence and pooled-variance working model; two-comparison Bonferroni family; explicit missingness limitation; bounded completion rule	Complete-record estimator and uncertainty aligned to contrast 1	Complete-record estimator and uncertainty aligned to contrast 2
STAT-I	Omnibus result retained as context; bounded completed-data stress test across 289 completions; the criterion is prospective only inside the hypothetical contract, while this application to existing data is post hoc	Complete-record estimate 3.09 points and interval [0.88, 5.30]; lower bound stays positive across bounded completions	Complete-record estimate 1.09 points and interval [-1.12, 3.30]; interval includes zero in every bounded completion

Statistical component	Shared core	Strand 1: DRTA minus DRA	Strand 2: Think-aloud minus DRA
STAT-C	The joint criterion remains unmet under the complete-record model and every scale-bounded completion; this is not identification of the all-assigned effects	Positive for the directional criterion under the stated calculations	Inconclusive for the directional criterion under the stated calculations

The shared source facts are **stated in the paper** or **located in related materials**. The exact package values are located in the public *carData* record. The teaching question, two estimands, Bonferroni family, and integrated rule are **chapter-created analytic content**, not statuses assigned to a source decision. A historical trace showing that Baumann and colleagues used this exact contract is **not located**; no conflict is claimed.

Table 11
Substantive-layer reading record for the integrated teaching argument.

Substantive component	Content identified during reading
SUB-E	Fourth-grade students in one rural Midwestern school; three instructional assignments; error detection as one 0–16 outcome
SUB-J	The randomized comparison contextualizes assignment effects in the documented setting; the completion stress test does not supply the missingness mechanism, implementation, educational importance, or wider relevance
SUB-I	Interpret both strand results together: one positive comparison cannot erase the other comparison's inconclusive uncertainty; stability of the rule across bounded completions is narrower than identification of effects
SUB-C	The statistical package criterion is not met; the integrated substantive answer is inconclusive and bounded to the selected outcome, pooled model, bounded completion exercise, documented setting, and post hoc application of the hypothetical contract

SUB-E situates the statistical evidence. SUB-J **augments and contextualizes the statistical justification**. SUB-I interprets the two statistical results without averaging one away. SUB-C carries their uncertainty and conditions into one bounded answer. The substantive layer cannot automatically strengthen weak statistical support.

4.5 Secondary use: evaluate quality

Quality evaluation begins only after the arguments have been read. Apply all six Chapter 1 questions without adding a score.

Table 12
All six quality-evaluation questions applied after reading.

Evaluation question	UCI single-strand pair	Hypothetical two-strand pair
Question alignment	**Adequate for this purpose:** event, denominator, estimand, estimate, and local answer match.	**Adequate for this purpose:** both contrasts and the conjunctive rule address the one integrated question.
Evidence integrity	**Needs revision** for broader use: selection, possible within-school dependence, and the zero-grade pattern are not fully documented.	**Needs revision:** two assigned outcomes are unavailable and only one selected outcome is analysed.
Justification adequacy	**Insufficient information to judge** a broad repeated-record process or a common meaning for the zero-grade cluster; the literal all-row description itself remains reproducible.	**Needs revision:** pooled variance and observed-to-assigned missingness support are limited; bounded completion is not a missingness mechanism.
Inferential adequacy	**Adequate for the stated, limited purpose:** magnitude, interval, omnibus test, and coding diagnostic are distinguished; substantive reasoning remains local and associational.	**Adequate for the stated, limited purpose:** the complete-record model calibrates the family and the bounded completion exercise stress-tests its decision; neither identifies the all-assigned effects.
Transition fidelity	**Adequate for this purpose:** the substantive interpretation retains magnitude, model status, and observational scope.	**Adequate for this purpose:** both intervals, the inconclusive strand, the post hoc application to existing data, and missingness survive the transition.
Claim calibration	**Adequate for this purpose:** the answer is local and associational.	**Adequate for this purpose:** the answer says the criterion is not established, not that the package failed or that either method is ineffective.

For the bounded answers as written in this chapter, the whole-pair evaluation conclusion is **supported as stated** for each pair. That conclusion does not mean every underlying study feature is strong. It means each reported answer already preserves the limitations exposed by the per-question judgments. In particular, an inconclusive answer can be supported as stated. Failure to meet a hypothetical success rule is not automatically the evaluation conclusion **not supported**.

Authors and analysts have the first responsibility to document and evaluate their own reasoning. Readers, collaborators, supervisors, reviewers, and editors can then apply the same questions independently.

4.6 Auxiliary use: construct the argument

The six-stage workflow supports conducting research. Here it is applied to how a **future hypothetical replication** could construct the integrated argument, not to infer an undocumented history for the 1992 study.

Table 13
Six-stage construction application for a hypothetical replication.

Phase and stage	Application to the hypothetical teaching contract	Status when applied to the existing record
Phase 1, Stage 1: Review problem, setting, and literature	Study the instructional context, outcome meaning, prior evidence, and feasibility of collecting complete follow-up	Source rationale is documented; the replication decision is chapter-created
Phase 1, Stage 2: Frame question and candidate answer	Declare one integrated question, both subordinate estimands, and an affirmative answer only if both criteria are met	Prospective only inside the hypothetical contract; post hoc relative to the available data
Phase 1, Stage 3: Align design, measurement, estimand, evidence plan, and inference	Align three-arm assignment, common 0–16 measure, complete-outcome procedures, zero benchmark, two-sided 95% Bonferroni intervals, and conjunctive rule	The chapter can demonstrate the analysis but cannot create historical prospectiveness
Phase 2, Stage 4: Produce or acquire and document evidence	Assign, deliver, measure, retain flow records, and document deviations	Existing source materials supply 68 assigned and 66 observed cases

Phase and stage	Application to the hypothetical teaching contract	Status when applied to the existing record
Phase 2, Stage 5: Estimate, check, and analyse	Inspect raw groups; estimate both complete-record contrasts; compute simultaneous intervals; keep the omnibus test in its contextual role; stress-test all 289 scale-bounded completions	Reproduced in Section 3
Phase 2, Stage 6: Contextualize, interpret, and complete the bounded answer	Retain both strands, missingness, analytic status, model conditions, completion-test limits, setting, and the inconclusive integrated result	Completed in the bounded answer

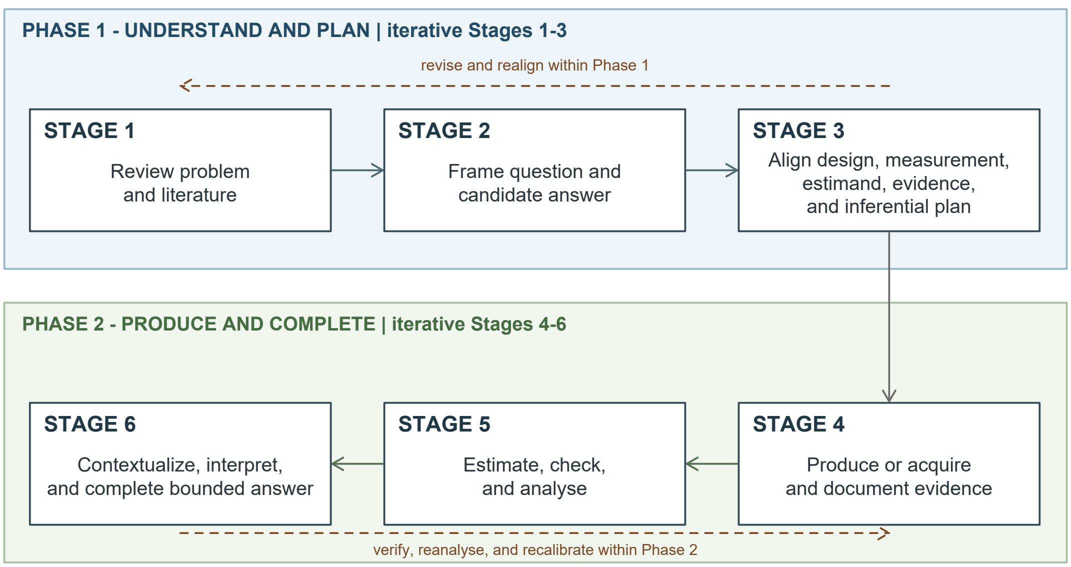

Figure 6: The six-stage construction workflow for the hypothetical replication.

Iteration stays within a phase. Rechecking a plot, correcting a table, or adding a same-estimand sensitivity analysis can revise Phase 2. Changing an estimand, outcome, design, or joint rule after seeing the evidence changes the Phase 1 anchor and starts an additional or revised cycle. Defensible material from the earlier cycle can be reused with its provenance and changed status visible; reuse cannot make a retrospective decision prospective.

5 Summary and Reading Bridge

5.1 A compact reading route

When reading a published group comparison:

1. Identify the unit, analytic sample, groups, outcome, estimand, and design.
2. For a categorical outcome, inspect counts and choose the denominator before reading percentages.
3. For a quantitative outcome, inspect raw grouped observations before reading ANOVA output.
4. State an effect estimate in interpretable units.
5. Use omnibus evidence only for its joint question.
6. Count statistical strands by jointly required estimands, not by output rows, coefficients, or tests.
7. If a result receives its own question-facing conclusion, give it a separate question–answer pair rather than hiding it inside another argument.
8. Carry uncertainty, analytic status, missingness, and design into the answer.
9. Read one argument with linked layers for each question–answer pair.

Reading is the primary use. Quality evaluation is the distinct secondary use after reading; the construction workflow is the auxiliary use for conducting research.

5.2 Compact glossary

Analytic sample. The observations that actually enter a reported analysis.

Conditional proportion. A proportion whose denominator is restricted to a declared group.

Estimand. The precisely specified quantity a study seeks to learn. An estimate is the numerical result obtained from observed evidence for it.

Expected count. A cell count implied by a stated null pattern, such as independence, given the observed margins.

Omnibus question. A joint question about a complete pattern, such as whether all declared group means are equal.

Statistical strand. An estimand-specific statistical chain inside one argument, used when sibling estimands are jointly required for one integrated answer and none is owed a separate conclusion.

Familywise calibration. An uncertainty or testing procedure that accounts for a declared family of comparisons.

Design-bounded claim. An answer whose wording retains the support and limitations of sampling, assignment, measurement, analysis, and context.

Inconclusive answer. An answer stating that materially different possibilities remain compatible with the evidence under a relevant benchmark and the declared conditions. It is not proof of no difference.

5.3 Reading bridge

These public sources extend the chapter without becoming prerequisites:

- Introduction to Modern Statistics: Exploring categorical data develops counts, proportions, and displays.
- Introduction to Modern Statistics: Inference for two-way tables develops expected counts and chi-square reasoning.
- Introduction to Modern Statistics: Inference for many means develops the omnibus ANOVA question and follow-up comparisons.
- NIST: Assessing a response with contrasts connects group means to declared contrasts; its Bonferroni guide explains simultaneous intervals for a finite comparison family.
- FDA guidance on multiple endpoints supplies a limited analogy for a conclusion that requires every declared component; it is not imported here as a social-science regulatory rule.
- UCI Student Performance record documents the observational source used here.
- The carData package documents the public Baumann record. The original report supplies the design context (Baumann et al., 1992).

Exercises

The exercises contain no answers and require no software. Use only **Foundational** and **Integrative** as levels. Keep the question, denominator, estimate, uncertainty, and design boundary visible in every response.

Exercise 1: Choose the denominator (Foundational)

A study records the following outcomes:

Participation format	Completed	Did not complete	Total
Online	84	36	120
In person	63	17	80
Total	**147**	**53**	**200**

The question is: "Among participants using each format, how did completion differ?"

1. Calculate the completion proportion within each format.
2. Calculate the proportion using the online format within each outcome category: completed and did not complete.
3. Identify which pair of proportions answers the stated question and explain why the other pair answers a different question.
4. State what the four raw counts contribute that percentages alone conceal.

Exercise 2: Name the categorical effect (Foundational)

Use the participation-format table in Exercise 1.

1. Calculate the Online-minus-In-person difference in completion proportions.
2. Interpret its sign, direction, event, and unit in one sentence.
3. Explain how the numerical value and wording would change if the event were changed from completion to noncompletion.
4. State one reason a proportion difference is easier to interpret here than an odds ratio.

Exercise 3: Check expected counts and chi-square output (Foundational)

For the participation-format table, the expected counts under independence are:

Participation format	Completed	Did not complete
Online	88.2	31.8
In person	58.8	21.2

Pearson's result is $\chi^2(1) = 1.89$, $p = .170$.

1. Use the observed table in Exercise 1 to reproduce the Online/Completed expected count, its contribution $(O-E)^2/E$, and the table degrees of freedom. Explain what an expected count represents.
2. Identify the cells in which the observed count exceeds the expected count.
3. State the omnibus null in words.
4. Write one conclusion that distinguishes the observed proportion difference, evidence against independence, association magnitude, and causation.
5. Explain why $p = .170$ is not proof that format and completion are independent.

Exercise 4: Connect grouped observations to ANOVA (Foundational)

Three groups have the following observed scores:

- Group A: 4, 5, 6, 6, 7, 8, 9, 9
- Group B: 5, 6, 7, 7, 8, 9, 10, 10
- Group C: 1, 4, 6, 8, 10, 12, 14, 17

Before considering an ANOVA result:

1. Compare the centers, spreads, overlap, and ranges.
2. Identify which group makes a common-variance model most questionable and explain why.
3. Explain what an individual-observation plot would reveal that a bar chart of means would conceal.
4. State why these observations cannot by themselves determine whether a group difference is causal or generalizable.
5. A one-way calculation gives $SS_B = 20.333333$ and $SS_W = 245.000000$. With three groups and 24 scores, calculate the between-group and within-group degrees of freedom, both mean squares, and F. Explain what the ratio compares; do not infer a specific pairwise difference from it.

Exercise 5: Distinguish outputs, strands, and pairs (Integrative)

Classify each situation as **several outputs in one strand**, **two statistical strands in one argument**, or **two question–answer pairs and arguments**. Explain the answer without counting table rows or hypotheses mechanically.

1. A report gives a mean difference, standard error, confidence interval, p-value, and sensitivity analysis for the same estimand.
2. One prespecified package-success question requires two declared estimands to meet their criteria; neither receives a separate conclusion.
3. A report makes one recommendation about Method A versus control and a separate recommendation about Method B versus control.
4. An omnibus ANOVA and a contrast interval are printed for one estimand- focused question. Is the omnibus result automatically another strand?

Exercise 6: Construct and read a two-strand result (Integrative)

A hypothetical teaching contract requires the lower endpoint of each two-sided 95% Bonferroni simultaneous interval to be above zero. Its two comparisons are Method B minus reference and Method C minus reference. Use these observed summaries:

Group	Mean score	Observed n
Reference	6.681818	22
Method B	9.772727	22
Method C	7.772727	22

Assume the independent, common-variance Normal model for this calculation. The pooled within-group mean square is $MS_W = 10.166667$ with 63 degrees of freedom. For this family of two contrasts, the supplied critical value is $t^* = 2.296$; no software or quantile lookup is needed. In group order reference, B, C, the weights are $(-1, 1, 0)$ and $(-1, 0, 1)$.

1. Calculate both contrast estimates and use $SE = \sqrt{MS_W(1/n_{\text{method}} + 1/n_{\text{reference}})}$ to calculate their common standard error.
2. Form each interval as estimate $\pm 2.296(SE)$, round endpoints to two decimal places, and state each strand's result in ordinary language.
3. Does the integrated statistical criterion succeed? Distinguish that decision from the substantive conclusion, using the declared zero benchmark.
4. Explain why neither the omnibus ANOVA nor the average of the two estimates can replace either required comparison.
5. Write one sentence distinguishing **inconclusive** from evidence that Method C failed or had exactly no effect. Name one design or missingness condition that the calculation alone cannot establish.

Exercise 7: Repair two group-comparison claims (Integrative)

Repair each claim using the evidence in this chapter.

> Higher-education intention caused students to pass because the chi-square test was significant.

> Both innovative instructional methods improve reading achievement for all fourth-graders.

For each claim:

1. identify at least three words or omissions that exceed the evidence;
2. retain the relevant magnitude and uncertainty;
3. retain the analytic sample, design, hypothetical status, and missingness where relevant; and
4. write one bounded replacement answer.

Exercise 8: Read, evaluate, then trace construction (Integrative)

Use the hypothetical two-strand teaching argument in this chapter.

1. Identify shared STAT-E and STAT-J content, each strand's STAT-I and STAT-C, and the one SUB-E/J/I/C chain.
2. Give a documentary status for the source assignment counts, the package scores, and the chapter-created integrated criterion. Explain why the last item must not be attributed to the source investigators.
3. Apply all six quality-evaluation questions. Keep documentary status, per-question judgments, and the whole-pair evaluation conclusion distinct.
4. Classify correcting a mislabeled plot as a Phase 2 revision. Then explain why changing an estimand or the joint rule after seeing results requires an additional or revised cycle.
5. Identify evidence that could be reused in that cycle and explain why reuse cannot make a retrospective choice prospective.

Appendix A: Further Categorical Comparisons

Difference, ratio, and odds scales

For event probabilities p_1 and p_0:

$$\text{difference} = p_1 - p_0, \qquad \text{ratio} = \frac{p_1}{p_0}, \qquad \text{odds ratio} = \frac{p_1/(1-p_1)}{p_0/(1-p_0)}.$$

Table 14

Three descriptive scales for passing in the UCI record

Scale	Estimate	Reading
Proportion difference	0.338	Percentage-point separation
Proportion ratio	1.97	Ratio of passing probabilities
Odds ratio	4.10	Ratio of passing odds

The scales are not interchangeable. Passing is common in the intention-yes group, so the odds ratio lies farther from 1 than the probability ratio. Reversing the group order takes the reciprocal of a ratio and changes the sign of a difference. Reversing the event changes the substantive story and the sign of the difference. A report must name the event and direction.

Larger and sparse categorical tables

For an $R \times C$ table, Pearson's statistic remains an omnibus comparison of observed and expected counts. Standardized residuals help locate cells that contribute to the pattern, while Cramer's

$$V = \sqrt{\frac{X^2}{n \min(R-1, C-1)}}$$

gives a unitless whole-table description. Neither supplies causal evidence or a universal practical-importance threshold.

Sparse tables need more care than a single expected-count slogan. In the artificial table

Group	Event yes	Event no
A	1	9
B	7	3

each expected count is 4 or 6. The usual chi-square reference can be poor. Fisher's exact test gives an exact conditional probability calculation for fixed margins. It is not exact about sampling, measurement, independence, causation, or truth.

Appendix B: Further Mean Comparisons

A complete small-data decomposition

The following six artificial scores isolate the arithmetic without adding another study narrative: group A has 1, 3; B has 3, 5; and C has 5, 7. Their means are 2, 4, and 6, and the grand mean is 4. Thus

$$SS_B = 2(2-4)^2 + 2(4-4)^2 + 2(6-4)^2 = 16, \qquad SS_W = 6(1^2) = 6.$$

Every score is one point above or below its own group mean. Directly around the grand mean, $SS_T = 9+1+1+1+1+9 = 22 = 16+6$. The mean squares are $MS_B = 16/(3-1) = 8$ and $MS_W = 6/(6-3) = 2$, giving $F = 8/2 = 4$. This illustrates the calculation; six artificial scores do not establish that the Normal common-variance model is suitable for a real study.

The algebra behind the identity is equally short. Expanding the square of $(\bar{Y}_j - \bar{Y}) + (Y_{ij} - \bar{Y}_j)$ adds a cross-term. Within group j, that term sums to $2(\bar{Y}_j - \bar{Y})\sum_i (Y_{ij} - \bar{Y}_j) = 0$ because deviations from a group's own mean sum to zero. What remains is exactly $SS_B + SS_W$.

Simultaneous comparison families

The main text's Bonferroni intervals use

$$\widehat{\Delta}_k \pm t_{1-.05/(2g),63}\, SE(\widehat{\Delta}_k), \qquad g = 2.$$

The family contains exactly the two DRA-referenced contrasts. Bonferroni is transparent and valid for this declared finite family, but it is not uniquely required. A Dunnett procedure can use the shared-reference dependence more efficiently; Tukey instead calibrates all pairwise comparisons. The family must follow the question rather than a software default.

The two contrasts share the DRA mean. Under independent groups with common variance their covariance is

$$\mathrm{Cov}(\bar{Y}_2 - \bar{Y}_1, \bar{Y}_3 - \bar{Y}_1) = \mathrm{Var}(\bar{Y}_1) = \frac{\sigma^2}{n_1}.$$

Terms involving means from different groups have zero covariance; the shared reference contributes positively. Bonferroni remains valid without independence between contrasts because the probability that either interval misses cannot exceed the sum of their individual miss probabilities. This guarantee still requires each component interval's model justification.

Table 15

Bonferroni simultaneous intervals for the two declared teaching contrasts

Comparison	Estimate	SE	Lower	Upper
DRTA minus DRA	3.09	0.96	0.88	5.30
Think-aloud minus DRA	1.09	0.96	-1.12	3.30

Table 16

Tukey 95% familywise intervals for observed Baumann outcomes

Comparison	Difference	Lower	Upper	Adjusted p
DRTA-DRA	3.09	0.78	5.40	0.006
Think-aloud-DRA	1.09	-1.22	3.40	0.497
Think-aloud-DRTA	-2.00	-4.31	0.31	0.102

The realized ANOVA summaries are $\eta^2 = SS_B/SS_T = .144$ and $\omega^2 = [SS_B - (J - 1)MS_W]/[SS_T + MS_W] = .116$. They describe how much observed score variation the fitted grouping accounts for under these formulas. They do not replace an outcome-scale contrast, define practical importance, or say what proportion of variation was caused by instruction.

Blocking within one comparison question

A blocking variable can improve precision when it records a feature related to the outcome and the assignment process respects the blocks. It does not create a second research question automatically. For one declared contrast estimand, an unadjusted group comparison and a block-adjusted comparison are alternative analyses of the **same** estimand, and therefore remain in one statistical strand, only when the adjustment preserves the declared units, contrast, target, and weighting. If adjustment changes the target or the weighting, it is not merely another analysis within the same strand.

Readers should ask:

- What was the unit of assignment, and what defined a block?
- Was assignment carried out within blocks or was the variable merely added later to a model?
- Does adjustment preserve the original estimand or change its weighting?
- Is uncertainty aligned with the actual assignment and dependence structure?

Blocking may strengthen precision or reveal sensitivity; the label itself is not design evidence. Two-way factorial questions, interactions, marginal weighting, and coordinated question sequences belong in Chapters 8–9.

References

Baumann, J. F., Seifert-Kessell, N., & Jones, L. A. (1992). Effect of think-aloud instruction on elementary students' comprehension monitoring abilities. *Journal of Reading Behavior*, *24*(2), 143–172. https://doi.org/10.1080/10862969209547770

Cortez, P. (2008). *Student performance* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. https://www.john-fox.ca/Companion/index.html

Fox, J., Weisberg, S., & Price, B. (2022). *carData: Companion to applied regression data sets* (Versions 3.0-5). Comprehensive R Archive Network. https://doi.org/10.32614/CRAN.package.carData